Statistics?

A field that deals with collection, organization, analysis, interpretation and presentation of the data.

↳ for decision making

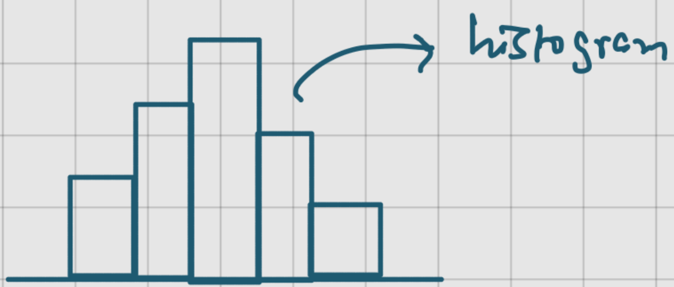
eg: Age = { 24, 27, 13, 14, 31, 32 }

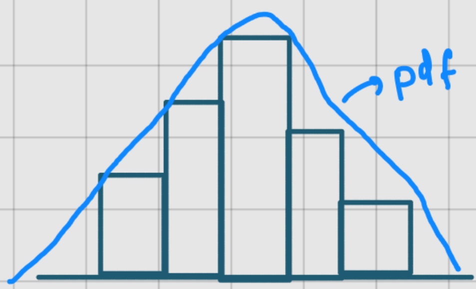
↓

feature ⟶ to target what to promote for what age group.

Charts & Graphs

1) Histograms
2) PDF   probability Density function
3) CDF   Cumulative Distribution function





Applications

i) ML
ii) DS
iii) Data Analytics
iv) Business Analyst
v) Risk Analyst

Types of statistics?

Statistics

Descriptive   Inferential

Descriptive:

- → Organising and summarizing of data
- → measure of central tendency
  - i) mean
  - ii) median
  - iii) mode
- → measure of Dispersion
  - i) variance
  - ii) Standard Deviation

Inferential

z test, t test

collect data using some experiments

↓

Conclusions or Inference

↓

other data → population data.

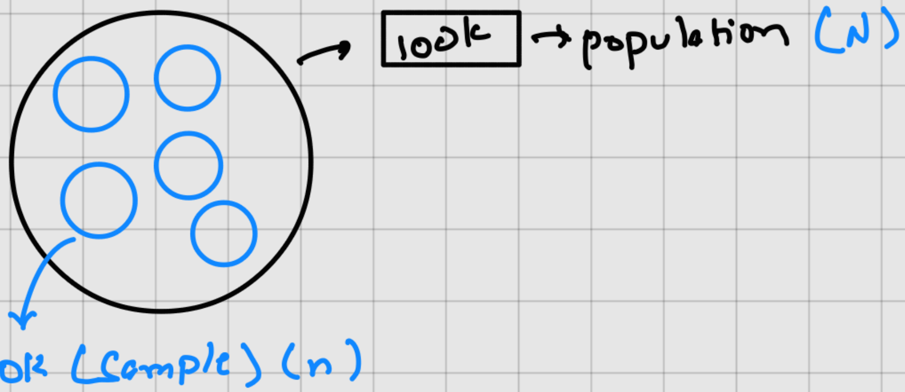
eg: sample set = 1000 students

heights = {180, 175, 162, 150, 160}

descriptive : mean, median   Avg = 165

inferential :

population vs sample data.



population	sample data
→ entire group	→ a subset taken from population
→ includes every possible observation	→ used to estimate population characteristics
→ usually large, expensive, impossible to fully measure	→ should be random and representative
∴ population = Truth, but often unreachable	∴ sample = approximation of truth
size : N	n
sd : σ	s
variance : σ^2	s^2
mean : μ	$\bar{x}$

Measure of central tendency: → where data lies

	N	n
i) mean: Symmetric data	$\mu = \sum_{i=1}^{N} \frac{x_i}{N}$	$\bar{x} = \sum_{i=1}^{n} \frac{x_i}{n}$

ages = {1, 3, 4, 5} → $\frac{1+3+4+5}{4}$ = 3.25

ii) median: (mid value): {1, 4, 3, 2, 5} → {1, 2, 3, 4, 5} median

skewed data / outliers in order

iii) mode: (max frequent value): {1, 2, 2, 3, 4}

$\underbrace{2,2}_{2}$ → mode

categorical / discrete

Measure of Dispersion → how spread out the data is

i) Range → max - min

ii) variance

iii) standard deviation → sqrt (variance)

iv) Quartile deviation

v) mean deviation

variance:

Age1 = {2, 2, 4, 4}

$\mu = \frac{2+2+4+4}{4}$

= 3

less spread

Age2 = {1, 1, 5, 5}

$\mu = \frac{1+1+5+5}{4}$

= 3

more spread

population data {N}

Sample data {n}

$$\sigma^2 = \sum_{i=1}^{N} \frac{(x_i - \mu)^2}{N}$$

Ages1 = {2, 2, 4, 4}

Ages2 = {1, 1, 5, 5}

$\mu = 3$

x_i	μ	$(x_i - \mu)^2$
2	3	1
2	3	1
4	3	1
4	3	1
	$\Sigma =$	4

$\frac{\Sigma}{n} = 4/4 = 1$

$\sigma^2 = 1$ (variance)

$sd = \sqrt{\sigma^2}$
$= 1$

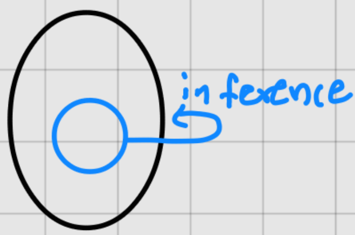
$\mu = 3$

$$s^2 = \sum_{i=1}^{n} \frac{(x_i - \bar{x})^2}{n-1}$$ → bassel correction

bessel

Why sample variance divided by n-1?

$\bar{x}$ = sample mean



$\bar{x} \approx \mu$

$s^2 \approx \sigma^2$

$\bar{x}$ is closer to sample points than true mean

The deviation $x_i - \bar{x}$ are slightly lower than $x_i - \mu$

* So dividing by n gives a number too small on average

we underestimate the true population variance (if we use n)

degree of freedom = $\boxed{n-1}$

Variables:

it is a property that can take any value.

eg: age = 10

age = [10, 20, 30] not a variable.

Variables

quantitative

qualitative / categorical

discrete

continuous

whole number

eg: 4, 3, 1, 10

(can exist only as a whole)

like number of bank account

can be floating (decimal)

eg:- 10.3, 4.5

like: balance in bank account

eg:- Colour — Red, blue

gender — male, female

(not a number)

Random Variables:

X

it is a value that is derived from a function or experiment

eg:- $y = 5x + 2$
$(x = 1, 2, 3, \dots)$

types

discrete

Continuous

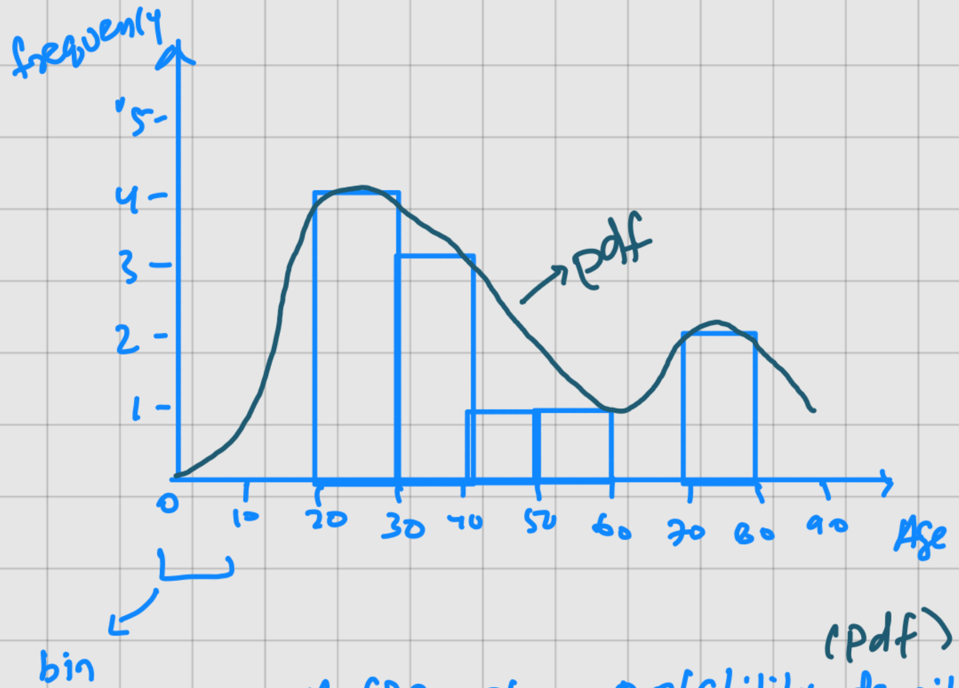
eg:- tossing a coin
rolling a dice

eg: amount of rain it'll rain tomorrow.
0, 1, 1.5, 1.55, . . .
age of people attending concert

Descriptive Statistics :-

Histograms : (counting based on bins)
used to derive even **probability density function**
using **kernel density estimation**

$x = \{ 23, 24, 25, 30, 34, 36, 40, 50, 60, 75, 86 \}$



20-30 = 4
31 - 40 = 3
41-50 = 1
51-60 = 1
61-70 = 0
71-80 = 2

(pdf)
we can get a probability density function if we smoothen the histogram.

But? how do we smoothen it?

By using **kernel density estimator**

Percentiles & Quartiles.

Percentage : { 1, 2, 3, 4, 5, 6 }
odd = 3 → 50%
even = 3 → 50%

Percentiles:

it is a value below which certain percentage of observations lie.

$\{2, 2, 3, 4, 5, 5, 6, 7, 8, 8, 9, 9, 10\}$

percentile of value x : $\frac{\text{\# of values below } x}{n} \times 100$

let $x = 9$. $\Rightarrow$ $\frac{11}{14} \times 100$

$\Rightarrow$ 78.57 percentile of value 9.

(ie) 78.57 % of the values are less than 9.

(2) $\text{value} = \frac{\text{Percentile}}{100} \times (n+1)$

$\Rightarrow \frac{\cancel{25}}{\cancel{100}} \times \cancel{15}$ $\underline{\underline{3.75}}$ $\longrightarrow$ $\frac{3+4}{2} = \underline{\underline{3.5}}$

25 percentile = 3.5

Quartiles:

25 % = 1^{st} quartile

50 % = 2^{nd} quartile

75 % = 3^{rd} quartile

every 25 % is a quartile.

5 Number Summary

$(Q_3 - Q_1)$
IQR :- inter quartile range

i) minimum
ii) 1^{st} Quartile (25%ile)
iii) median
iv) 3^{rd} quartile (75%ile)
v) maximum

lower fence = $Q_1 - 1.5(IQR)$
higher fence = $Q_3 + 1.5(IQR)$

anything lower than lower & anything higher than higher fence is removed

eg :- 1, 2, 2, 2, 3, 3, 4, 5, 5, 5, 6, 6, 6, 6, 7, 8, 8, 9, ~~27~~ (removed)

→ outlier

↳ cuz greater than higher fence

i) 1 (min)
ii) 3 (5^{th} position) (25%)

$$\frac{1}{4}\frac{25}{100} \times (19+1)$$

$$\frac{1}{4}(20)$$

iii) 5.5 (median)

iv) 7 (15^{th} position) (75%)

$$\frac{3}{4}\frac{75}{100} \times (20)$$

15

v) 27 (max)

IQR = 7 − 3 = 4
lowerfence = $Q_1 - 1.5(IQR)$
$3 - 1.5(4)$
$= 3 - 6$
$= -3$

higher fence = $Q3 + 1.5(IQR)$

$7 + 1.5(4)$

$7 + 6$

$= \underline{\underline{13}}$

After removing outliers:

1, 2, 2, 2, 3, 3, 4, 5, 5, 5, 6, 6, 6, 6, 7, 8, 8, 9

min = 1
max = 9
1st quartile = 3
median = 5
3rd quartile = 7

$\Bigg\}$ five number summary

↓

Box plot

whisker plot

Correlation and Covariance.

Covariance and Correlation are two statistical measurements used to determine the relationship between two variables

Both are used to understand how changes in one variable are associated with changes in another variable.

Covariance:

It is a measure of how two random variables change together.

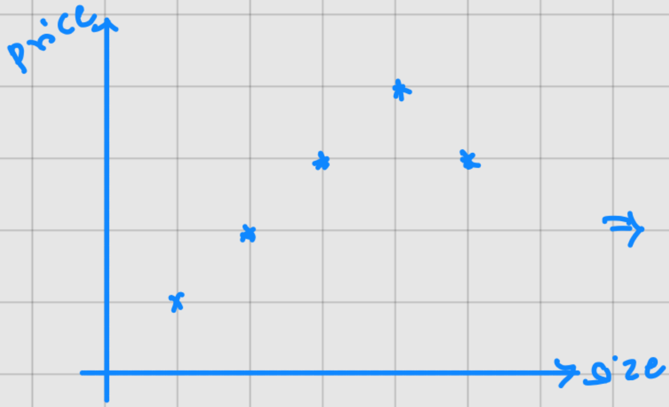
→ if the variables tend to increase and decrease together then covariance is positive.

→ if one tends to increase and other tends to decrease then the covariance is negative.

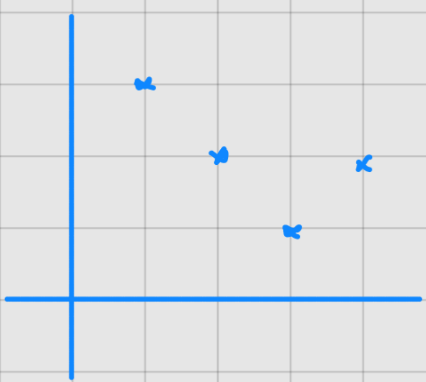
Data set

Size of house	Price
1200	45 L
1300	50 L
1500	75 L
1100	40 L

Size increases, price also increases and vice versa



⇒ positive covariance



⇒ negative covariance

$$\text{Covariance}(x,y) = \sum_{i=1}^{n} \frac{(x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

$$\text{covariance}(x,x) = \sum_{i=1}^{n} \frac{(x_i-\bar{x})(x_i-\bar{x})}{n} \Rightarrow \underbrace{\sum_{i=1}^{n} \frac{(x_i-\bar{x})^2}{n}}_{\text{variance of } x}$$

x_i = Data points of random variable x

$\bar{x}$ = sample mean of x

y_i = Data points of random variable y

$\bar{y}$ = Sample mean of y

Students data set

Hours studied (x)	Exam score (y)
2	50
3	60
4	70
5	80
6	90

$x\uparrow\ y\uparrow$

$x\downarrow\ y\downarrow$

① $\bar{x} = \frac{(2+3+4+5+6)}{5} = 4$ $\qquad \bar{y} = \frac{(50+60+70+80+90)}{5} = 70$

$$\text{Covariance}(x,y) = \frac{(2-4)(50-70)+(3-4)(60-70)+(4-4)(70-70)+(5-4)(80-70)+(6-4)(90-70)}{5}$$

$$= \frac{(-2)(-20)+(-1)(-10)+(0)+(1)(10)+(2)(20)}{5}$$

$$= \frac{40+10+10+40}{5}$$

$$= \frac{100}{5}$$

$= 20 \rightarrow$ positive value so positive cov positive covariance $\therefore$ no of hours studied increases the score

Advantages :

i) quantify relationship between x and y

Disadvantage:

i) does not have a specific limit value $(-\infty, \infty)$
(so we use co-relation)

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② Co-relation

Correlation

pearson correlation coefficient   Spearman rank co-efficient

i) pearson correlation coefficient [-1 to 1]

$$\rho_{x,y} = \frac{Cov(x,y)}{\sigma_x \sigma_y}$$

The more the value towards +1, the more +ve correlation of x & y

The more the value towards -1, the more -ve correlated x & y

pearson correlation cannot capture correlation for non-linear data

↓

Solved by Spearman rank correlation.

ii) Spearman rank correlation.

$$r_s = \frac{\mathrm{cov}(R(x), R(y))}{\sigma(R(x)) * \sigma(R(y))}$$

x	y	R(x)	R(y)
1	2	2	1
3	4	3	2
5	6	4	3
7	8	5	5
0	7	1	4

feature selection.

haunted

size of house ↑ no. of rooms ↑ location ↑ price

-ve correlated

+ve correlated

~~no. of people~~

not necessary

Skipped